Paleomagnetic Study of Satyavedu Sandstones of Cretaceous Age from Andhra Pradesh, India*

By G. S. MITAL [1]), R. K. VERMA [2]) and G. PULLAIAH [3])

Summary – The paleomagnetism of twenty six oriented samples of red sandstones from three different sites located in Satyavedu hills (near 13°30′N, 79°55′E) and belonging to coastal upper Gondwana formations of India has been studied. These sandstones have been considered to be equivalent of Tirupati sandstones of Lower to Middle Cretaceous age from Godavari valley on stratigraphic considerations. Consistent directions of magnetization were obtained from two different sites after stability tests. Results from one site gave ancient pole position at 26°N, 67°W, very close to that obtained for Tirupati sandstones, thus confirming the geologic correlation. Results from the second site were rejected on account of instability and those from the third site gave pole position at 79°48′N, 76°58′W. This site appears to have been remagnetized during Pleistocene times when there was large deposition of laterite in the area. Study of magnetic properties of these sandstones revealed that the magnetization was of the nature of CRM and the NRM was carried almost entirely by red coating on silica grains.

Introduction

Detailed paleomagnetic studies of several Mesozoic formations in India have proved to be quite useful for studying the continental drift of the Indian landmass. The data has been recently summarised by ATHAVALE *et al.* [1]) [4]. Amongst the Mesozoic formations, the Deccan traps ranging in age from Upper Cretaceous to Eocene have been studied in great detail by BHIMSHANKARAM and PAUL [2], CLEGG *et al.* [4] DEUTSCH *et al.* [9], VERMA and PULLAIAH [17] and SAHASRABUDHE [18]. One sedimentary formation of Cretaceous age which has been studied in detail is Tirupati sandstone formation of Lower to Middle Cretaceous age from Godavari valley [15, 16]. The results from this formation have given pole position very close to that obtained for lower Deccan traps (of slightly younger age) and thus provide an independent confirmation of paleomagnetic results. There is however always some uncertainty regarding the age of magnetization of the red sandstones and hence it is desirable to obtain further evidence regarding the continental drift and the nature of geomagnetic field during the geologic past by studying contemporaneous formations from different parts of the country.

The Cretaceous period in India is characterised by sedimentation at several differ-

* N.G.R.I. Contribution No. 70–169.

[1]) National Geophysical Research Institute, Hyderabad-7, India.

[2]) Present Address – Indian School of Mines, Dhanbad, India.

[3]) Present Address – Air Borne Mineral Surveys and Exploration, New Delhi, India.

[4]) Numbers in brackets refer to References, pages 190/191.

ent places along the east coast. These formations also contain red sandstones and shales within their horizons. One of the well known exposures of Cretaceous age along the east coast is at Satyavedu hills in Chittoor district of Andhra Pradesh about 35 miles NW of Madras. Paleomagnetic studies on 26 samples of red sandstones from these exposures have given interesting results from the point of view of geological correlation using paleomagnetic methods. The results are discussed in this paper.

Geology of the area

The Upper Gondwana formations along the coastal districts of Madras have been studied by R. B. FOOTE [10]. There are four principle exposures of Upper Gondwanas namely, Satyavedu, Sripermatur, Alicoor and Pyanur. The lower stage named Sripermatur 25 miles WNW of Madras contains marine animal and plant remains. The Upper stage forming the Satyavedu beds consists of coarse, compact, purple mottled ferruginous sandstones and conglomerates and contains fragmentary plant fossils. The lower stage Sripermatur contains a number of plant fossils such as Cladeophlebis whitbyensis, C. indica, Thinnfeldia sp., Taeniopteris spatulata, Pagiophyllum peregrinum, Ginkgo crassipes (KRISHNAN [13]). These fossils are identical to those found in Raghavapuram shales. Stratigraphically Tirupati sandstones overlie the Raghavapuram shales in Godavari valley just as Satyavedu sandstones overlie the Sripermatur beds in Madras region. On account of this reason the Tirupati sandstones and Satyavedu sandstones are considered to be geologically equivalent. These are considered to be of Lower to Middle Cretaceous age as inferred by Spath (quoted by KRISHNAN, [13]). The Upper Gondwanas are surrounded by Laterite of Pleistocene age almost on all sides as indicated by geological map of the area in figure 1.

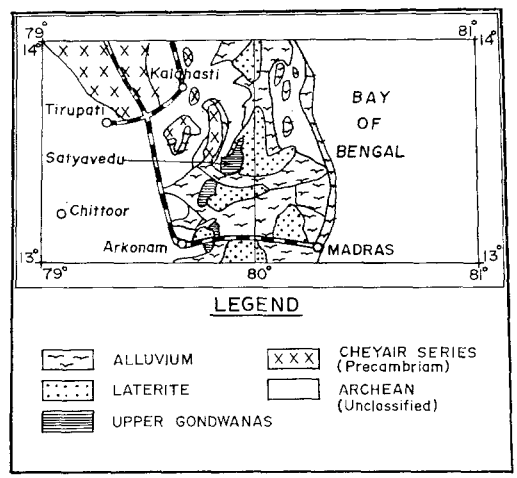


Figure 1
Geological map of the area showing the location of Satyavedu hills-geology after Geological Survey of India map, 1963

Sampling

Twenty six oriented samples of Satyavedu sandstones were collected from three different sites in quarries located at 13°30′N, 79°55′E (site SA); 13°24′N, 79°53′E (site SB) and 13°24′N, 79°53′E (site SC) in Satyavedu hills. The strike of the formation is N–S at all the sites while dips are gentle towards east. The sandstones are mostly coarse grained and reddish to brownish in color.

NRM Measurements

All measurements of remanent intensity were made using astatic magnetometer of sensitivity 3×10^{-7} Oe/cm deflection at 4 m. optical path. Out of each sample 4 to 6 specimens of 1″ diameter and 1″ length were studied. The NRM directions of three sites plotted on equal area projection are shown in figures 2a, 2b and 2c. The specimens from the same sample are joined together. The range of NRM intensities is fairly large. The mean values for three sites lie between 4.1×10^{-6} to 7.4×10^{-6} gauss. On the whole the intensities of specimens from site SA were much lower than those of sites SB and SC. In order to find any correlation of NRM intensity with magnetic properties, the magnetic susceptibility of each specimen was determined in a field of 0.5 Oe using magnetic susceptibility apparatus constructed in the institute

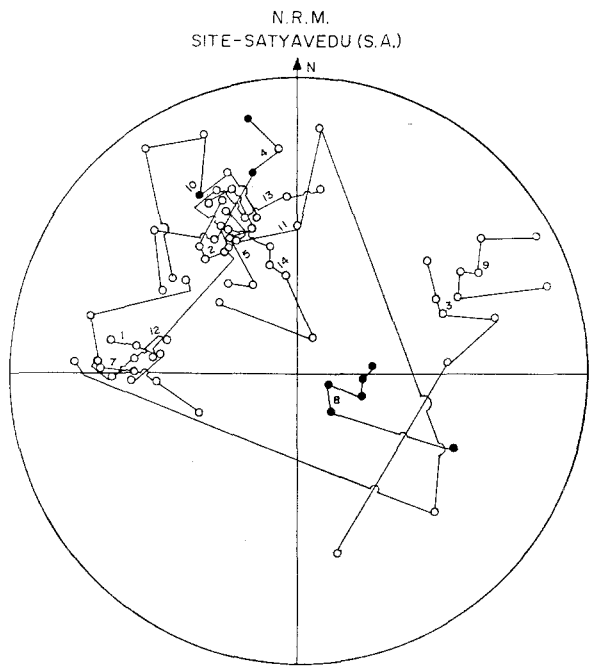


Figure 2a

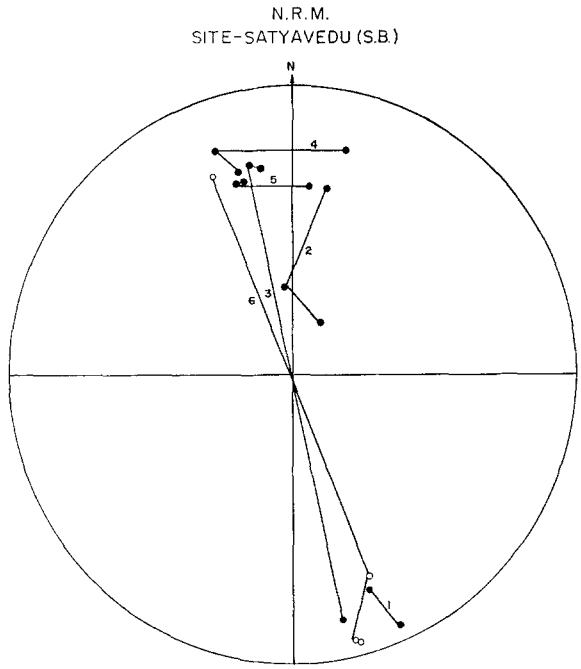


Figure 2b

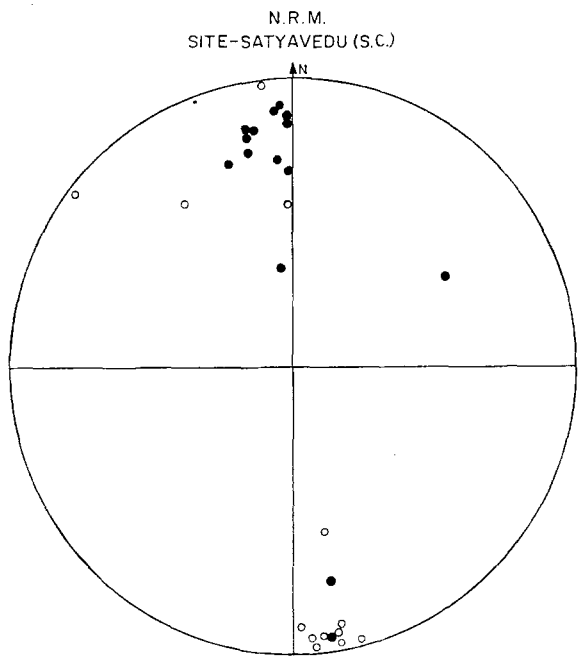


Figure 2c

NRM directions for sites SA, SB and SC plotted on equal area projection. Specimens from the same sample are joined • – down dip. ○ – up dip

Table 1

Range of NRM intensity, susceptibility and Q_n ratio for sites SA and SC

Site	NRM Intensity ($\times 10^{-6}$ gauss)		Susceptibility ($\times 10^{-6}$ gauss/oe)		Q_n Ratio	
	Range	Mean	Range	Mean	Range	Mean
SA	1.1–23.8	4.1	4.0–23.5	11.8	0.3–6.0	0.9
SC	0.4–32.1	7.4	3.4–14.6	5.5	0.2–7.7	3.4

after LIKHITE and RADHAKRISHNAMURTY [14]. The Q_n (Koenigsberger) ratios were determined from susceptibility values using the values of 0.5 Oe for intensity of earth's magnetic field. The range in values of the various parameters as well as mean values for NRM intensity, susceptibility and Q_n ratio for three sites are given in table 1. It was found that the specimens from site SA although having Lower NRM intensity as well as Q_n ratio showed stable directions which were quite different from those of sites SB and SC.

Stability tests

Site SA: Few selected specimens from this site were subjected to demagnetization in progressively increasing alternating fields upto 800 Oe peak value. It was found that the response of specimens to this technique was quite good but maximum clean-

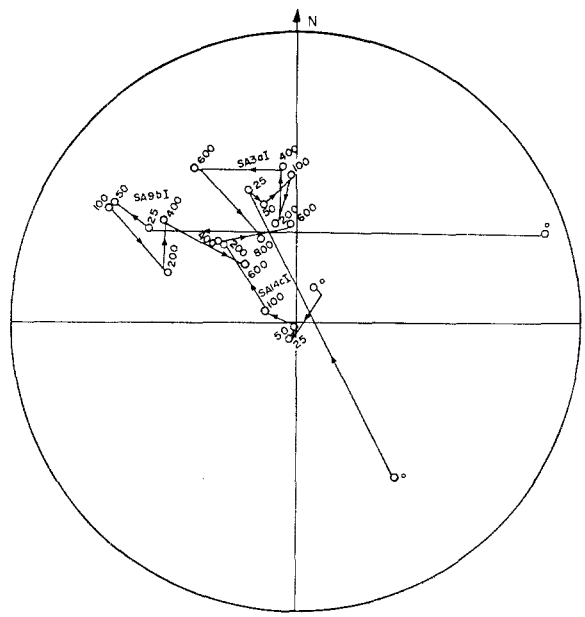


Figure 3

Change in the direction of magnetization of few typical specimens from site SA upon a.c. cleaning in successive steps of 25, 50, 100, 200, 400, 600 and 800 Oe peak fields

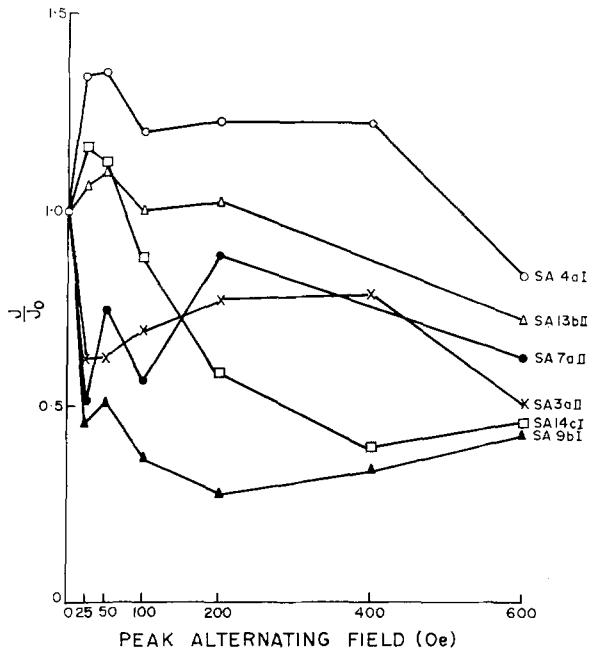


Figure 4
Normalized magnetic intensity vs peak alternating fields for few typical specimens from site SA

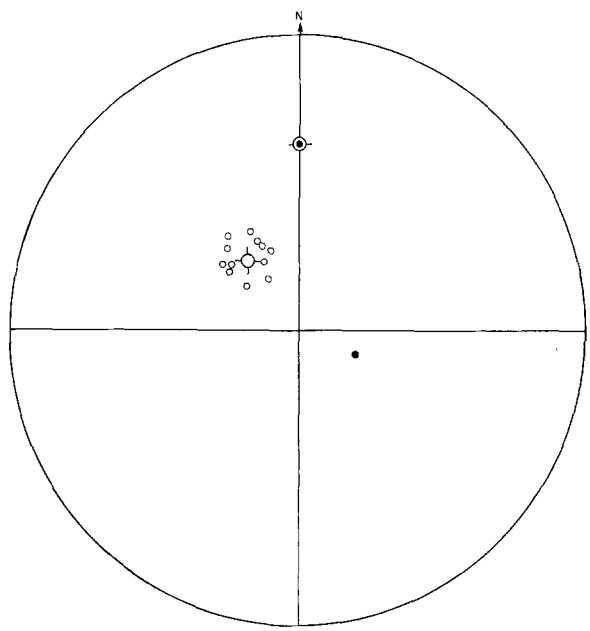


Figure 5
Mean directions of magnetization for samples from site SA after cleaning in peak alternating fields of 800 Oe. • – down dip, ○ – up dip, ⊹ – mean direction, ⊙ – present dipole field at site

ing was achieved in peak fields of about 800 Oe as was evidenced by maximum value of Fisher's precision parameter obtained. The rest of the specimens were cleaned in 400, 600 and 800 Oe peak fields. Few typical examples of change in vector upon a.c. cleaning are shown in figure 3. Maximum change in direction for specimens took place in fields upto 50 Oe peak value indicating the presence of viscous components. Alternating field demagnetization curves of normalized intensity for few representative specimens are shown in figure 4. The mean for each sample after demagnetization as well as mean for the site are shown in figure 5. The mean directions and other parameters for this site are given in table 3.

Site SB: The specimens from this site did not show any stability in direction after a.c. cleaning or on thermal demagnetization in sedimentary furnace. This site was therefore rejected.

Site SC: For this site the a.c. demagnetization technique was not very useful. Thermal demagnetization at progressively increasing temperatures was carried out in a sedimentary furnace placed in a field free space of about 8″ diameter. The earth's magnetic field was neutralized by using six sets of Helmholtz coil system designed after SCOTT [19]. The uniformity of the field in the space was upto 0.1%. The field could be neutralized upto 50 gamma using a milli oersted meter. Sufficient magnetic cleaning was achieved for all specimens in temperature of about 220 °C. The cleaned directions for all specimens show two well defined groups, about 180° apart as plotted in figure 6.

The mean directions for these are given in table 3 along with other paleomagnetic parameters. These directions are much different from those of site SA. Pole position for this site was calculated using both normal as well as reverse directions. Thermal demagnetization for this site was carried out upto 660 °C. The nature of thermal demagnetization curves is discussed later along with magnetic properties.

Nature of magnetization

It was considered desirable to make some tests which would reveal the nature of magnetization of these sandstones and reveal differences in their magnetic history. Although much of the evidence on continental drift and polar wandering is built upon the studies of red sandstones, the magnetic properties of these formations have not been discussed in sufficient detail. COLLINSON [5, 6, 7] has studied the magnetic properties of several red sandstones from Europe and North America.

It is generally considered that the NRM of red sandstones is carried by the red coating in the rock, which is a form of hydrated ferric oxide. COLLINSON [5] however has demonstrated that the NRM is not carried entirely by the red coating for all the formations (for instance Chugwater formation of Triassic age). Besides hydrated ferric oxide, specularite and titanohematites are suspected to be present in some of the red sandstones (COLLINSON [6]). Apparently the magnetic properties of red sandstones show a great deal of variation on account of their complex history of magnetization.

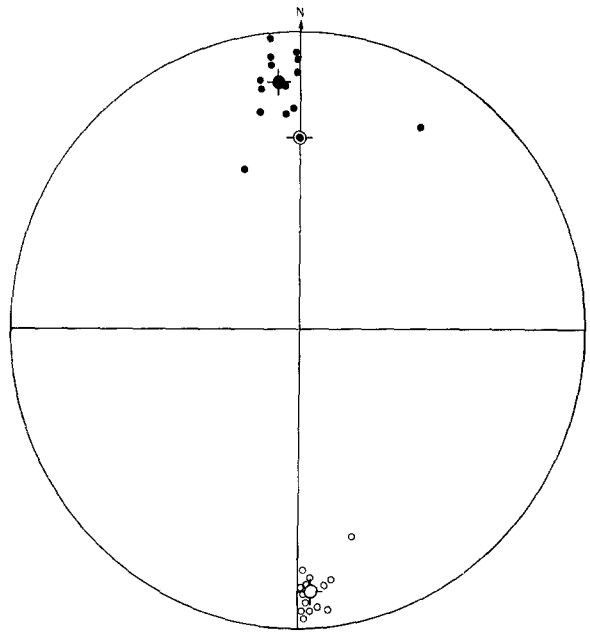


Figure 6
Thermally cleaned directions at 220 °C for site SC, ⊕ – mean for normal specimens, ⊖ – mean for reverse specimens, ⊕ – present dipole field at site

The studies which were made on these sandstones included thermal demagnetization of NRM, chemical demagnetization of NRM in a manner similar to that of COLLINSON [5] and remanent coercive force. Four samples from each of sites SA and SC were selected for these studies. The chemical analyses of these are given in table 2.

These analyses show that the chemical composition of the magnetic mineral is not pure hematite but is titanohematite. The chemical composition is plotted in a Ternary diagram in figure 7.

Thermal demagnetization curves of NRM intensity are shown in figure 8 for a few specimens from sites SA and SC. The nature of these curves suggests that these consist predominantly of distributed components which are most probably of chemical origin. These components may be associated with distributed blocking temperatures of domain walls which probably implies a wide range in grain sizes. Similar thermal demagnetization curves for red sandstones have been obtained by IRVING and OPDYKE [12] for Bloomsberg formation and by IRVING [11] for upper Kutung series from Australia.

For chemical demagnetization tests four discs (2.5–3.0 mm thickness) were prepared from sites SA and SC for which chemical analyses are given in table 2. The discs were quite porous and penetration of hydrochloric acid in each disc was quite good. The NRM of each disc was measured under the magnetometer after keeping it in acid for varying lengths of time 5, 10, 15, 40 min. etc. It was found that the

Table 2
Chemical composition of samples from sites SA and SC

Site	Specimen No.	(Wt. %) of			
		FeO	Fe_2O_3	TiO_2	Fe (Total)
SA	SA2	0.22	2.75	0.152	2.10
SA	SA3	0.14	2.98	0.227	2.20
SA	SA10	0.22	2.89	0.278	2.20
SA	SA11	0.22	2.61	0.253	2.00
SC	SC1	0.36	3.89	0.480	3.00
SC	SC3	0.22	3.75	0.505	2.80
SC	SC5	0.22	4.46	0.480	3.30
SC	SC7	0.14	4.55	0.505	3.30

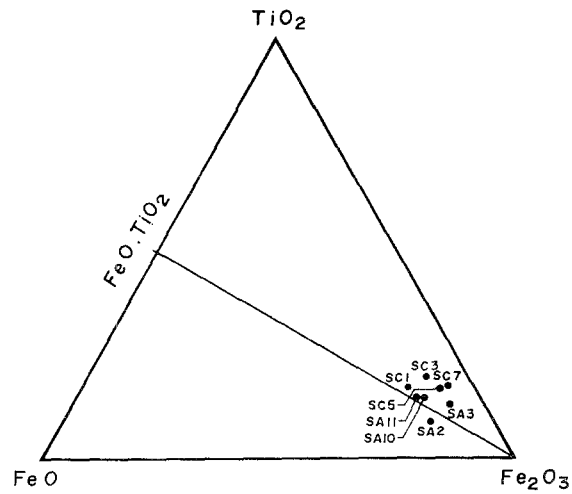


Figure 7
Chemical composition of the magnetic mineral plotted on ternary diagram

color of the disc started to change rapidly and became white subsequently. Figure 9 shows the curves of decay of NRM with time for a number of discs from both sites.

These curves are quite similar to those obtained for Supai formation of Permian age from Arizona, which is believed to have been remagnetized during Quaternary times [5]. After keeping the discs in the acid for a cumulative period of about three hours the color of the discs became almost white and the percentage of NRM left over was about 15–20%. This was mostly due to some reddish material remaining inside the disc as the time of penetration of hydrochloric acid deeper inside was longer. These discs (after washing) were magnetized in a field of 14,000 Oe to see if any other magnetic material was present in the discs in significant amount. The IRM acquired by the (acid) treated discs was compared with that of untreated discs of the same thickness. It was found that the ratio of IRM of the disc (after/before) treatment

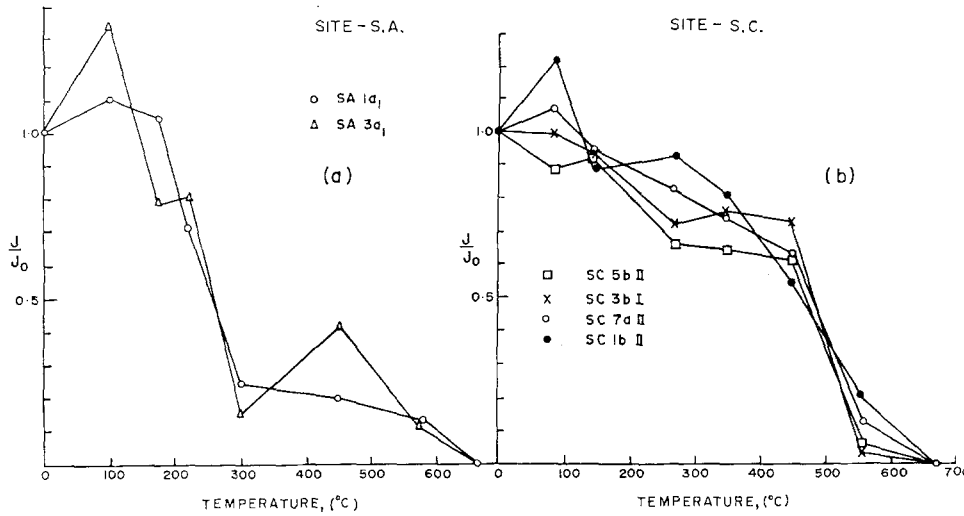


Figure 8
The Thermal demagnetization curves of normalized intensity for few typical specimens from sites SA and SC

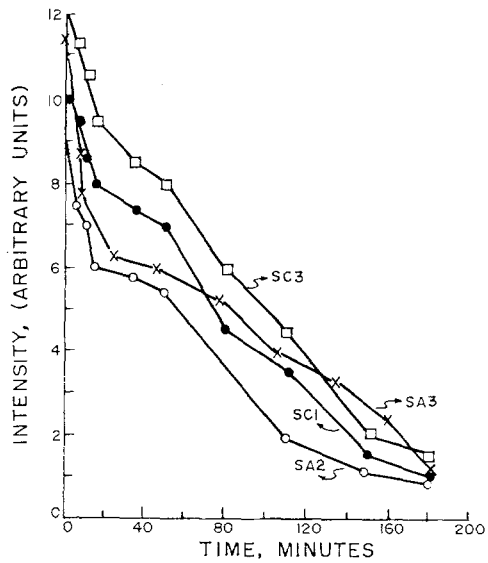


Figure 9
Decay of NRM intensity with time due to chemical demagnetization in conc. hydrochloric acid for typical specimens from sites SA and SC

was the same as the percentage of NRM. This showed that no other material besides the red coating was responsible for the NRM in these rocks and no significant amount of this material was present in unmagnetized state.

Coercivity of remanence is considered to be an important property of the mag-

netic mineral. COLLINSON [6], PULLAIAH and VERMA [15] have reported variation in this parameter from 1000 Oe to 6000 Oe for various red sandstone formations. The value of coercivity depends upon the chemical composition as well as the grain size. The coercivity values for several specimens from both the sites were determined in a manner similar to that of CARMICHAEL [3]. The partial hysteresis curves for two typical specimens from sites SA and SC are shown in figure 10. It is noticed that the

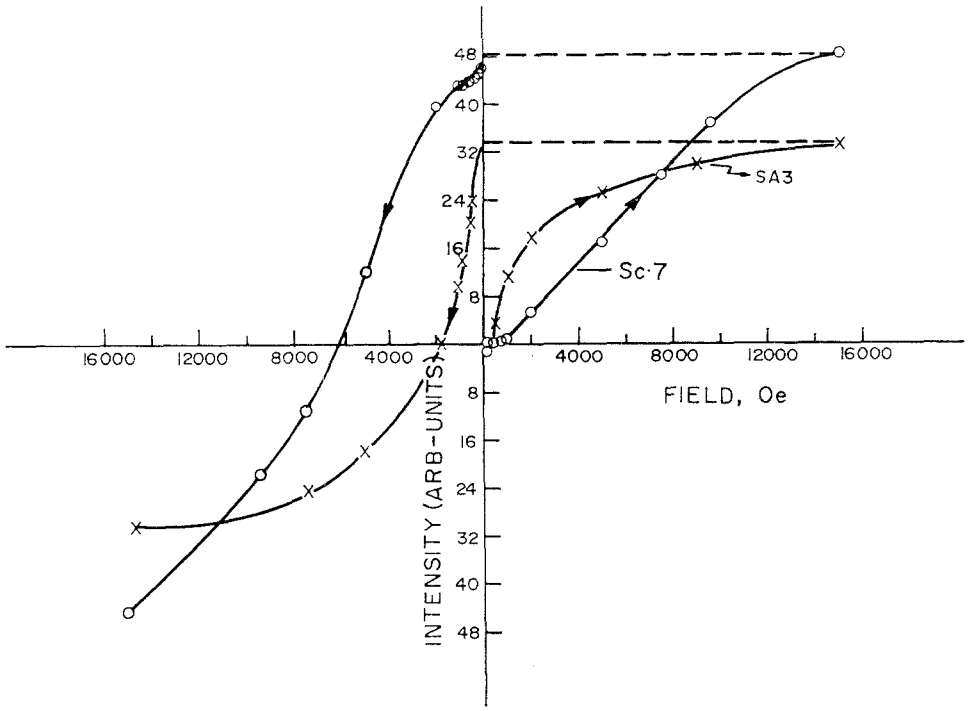


Figure 10
Partial hysteresis curves of remanence for two typical specimens from sites SA and SC

coercivity is appreciably lower for site SA (about 2000 Oe) as compared to that for site SC (about 6000 Oe). This difference may be due to different grain sizes in the samples from two sites.

The various magnetic properties discussed above showed that the magnetization of samples from both the sites was essentially due to the red coating. The difference in their thermal demagnetization curves being probably due to different blocking temperatures of domain walls. No particular magnetic property studied could be ascribed to remagnetization of site SC.

Results and discussion

Paleomagnetic parameters obtained for sites SA and SC which showed stable

directions of magnetization are given in Table 3. Virtual geomagnetic poles were calculated assuming geocentric axial dipole field. The pole positions as well as ovals of confidence at 95% probability level are plotted for two sites in figure 11 along with poles obtained for four sites of Tirupati sandstones by VERMA and PULLAIAH [17]. It is seen that the pole obtained for site SA is very close to that of three sites of Tirupati sandstones (T_2, T_3, T_4). This result supports the geologic correlation between the two formations based upon stratigraphic evidence and also suggests that the age of magnetization of Tirupati sandstones and of Satyavedu (site SA) was nearly the same. A very close agreement in pole positions between the two formations separated by a distance of about 600 km. also indicates that the geomagnetic field must have been predominantly an axial dipole field during the time of magnetization of these beds.

The pole position for site SC is quite far from that of site SA. Although the nature of magnetization of this site appears to be predominantly CRM, this site appears to

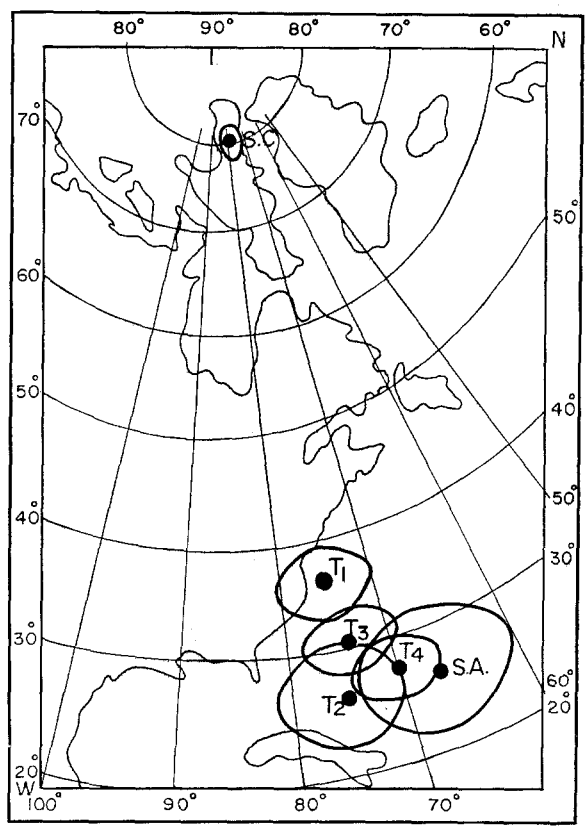


Figure 11
Virtual geomagnetic poles for sites SA and SC, of Satyavedu sandstones compared with those of Tirupati sandstones (T_1, T_2, T_3 and T_4) from four sites

Table 3

Paleomagnetic parameters for sites SA and SC

Site	Present								Virtual Geomagnetic pole					
No.	*N*	*n*	Lat.	Long.	*D*	*I*	*R*	*K*	α95	λ*p*	*Lp*	*dm*	*dp*	Polarity
SA	13	65	13°30′N	80°E	321°	−58°	12.85	80	4°23′	26°8′N	67°W	6.45	4.75	Normal
SC	7	28	13°24′N	79°53′E	355°	+11°	3.99	300	4°					Normal
										79°48′N	76°58′W	3.29	1.63	
					177°	− 7°	2.999	2000	2°					Reverse

N = Total Number of samples studied.
n = No. of specimens studied
D = Mean declination of stable remanence, East of North
I = Mean inclination of stable R.M., Down +ve and up −ve
R = Length of vector sum of *N* unit vectors
K = Precision Parameter
α95 = Radius of circle of confidence at 95% probability level
λ*p*, *Lp* = Latitude and longitude of V.G.P. (north pole)
dm, *dp* = Ovals of confidence about the virtual pole position.

have been remagnetized during later Tertiary times. From the geological map of the area it is seen that there is a wide spread deposition of laterite in the area during Pleistocene times. This site therefore appears to have been remagnetized during this period. As rocks from this site have shown normal as well as reverse polarity (almost 180° apart) this appears to correspond to geomagnetic field reversal during this period as inferred from studies of several igneous formations by Cox *et al.* [8]. The Paleo magnetic results from site SC however point out clearly one significant uncertainty associated with the study of red sandstones, the problem of determination of their exact age of magnetization.

Acknowledgement

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